INCH - POUND
A-A-56032A
10 March 1998
SUPERSEDING
A-A-56032
30 September 1996

COMMERCIAL ITEM DESCRIPTION

INK, MARKING, EPOXY BASE

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-I-43553B for all federal agencies.

1. SCOPE. This commercial item description covers a catalyzed epoxy system of marking inks for metallic or other non-porous surfaces and printed wiring boards.

2. CLASSIFICATION.

2.1 Type. The marking ink shall be of the following types as specified:

Type I - General usage
Type II - Nonconductive ink (for printed wiring boards)

3. SALIENT CHARACTERISTICS.

3.1 Materials. The Contractor is given latitude in the selection of raw materials and in manufacturing processes provided the product meets the required levels of quality and serviceability.

3.2 Color and quantity. The color and quantity of the cured ink shall be as specified in the purchase order or contract.

3.3 Composition. The ink shall be furnished in the form of two compounds - the basic ink

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compound and a hardener compound. The basic ink compound shall consist essentially of an epoxy resin in liquid, paste, or powder form. The hardener compound (catalyst) shall be in liquid, paste, or powder form for mixing with the ink compound just prior to use. The hardener shall cause the mixture to harden (cure) through chemical reaction at room ambient temperatures, or at other time/temperature ratios, as specified. The chemical composition of each compound shall be at the discretion of the manufacturer.

3.3.1 Additives. Ingredients may be added to the ink and hardener compounds to produce mat finishes. Other additives may be used to improve the leveling, flexibility, and adhesive qualities. No additives to Type II ink shall affect its electrical resistance requirements (see 3.8).

3.3.2 Thinner. When specified in the contract or purchase order, a non-toxic thinner shall be furnished for reducing the ink for optimum application consistency. The thinner shall be compatible with the wiring board substrate.

3.4 Mixing properties. The ink and hardener compound shall mix freely to produce a homogeneous mixture, free from skins, lumps, sediment, and precipitant materials, and shall be of a consistency suitable for the intended method of application. After mixing, the chemical components shall not gel during the specified pot life of the ink at room ambient temperature, and chemicals that may separate shall be readily dispersible.

3.5 Pot life. The useful life of the combined two-part ink when mixed as specified by the manufacturer shall be not less than four hours, unless otherwise specified by the manufacturer.

3.6 Curing. The ink, mixed and applied to a vertical or horizontal surface and cured at times and temperatures recommended by the manufacturer, shall be completely cured for testing purposes.

3.6.1 Cure for handling. The combined two-part ink, when mixed as specified by the manufacturer, shall be cured for handling when exposed to a temperature of 65.5ºC for 30 minutes.

3.7 Adhesion. Cured ink impressions on glass, steel, aluminum, copper, and plastic shall not deteriorate when subjected to trichloroethylene vapors at 86.5º to 88ºC for a period of not less than three minutes and not greater than six minutes.

3.8 Electrical resistance (Type II only). The material cured, as specified herein, and prior to high humidity conditioning, shall have an electrical resistance of not less than 10^{12} ohms. After conditioning, the material shall have an electrical resistance of not less than 10^{10} ohms.

3.9 Environmental and storage conditions.

3.9.1 Abrasion resistance. Cured ink impressions shall retain their legibility after subjection to 300 to 303 revolutions of the CS-10 abrasive wheel while under a minimum load of 2.2 pounds.

3.9.2 Chemical resistance. Cured ink impressions shall not deteriorate when immersed for a

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minimum of 30 minutes in water, denatured ethyl alcohol, methyl alcohol, and cleaning solvent.

3.9.2.1 Chemical resistance (Type II only). In addition to 3.9.2, Type II cured ink shall be resistant to hot solder and solder flux.

3.9.3 Salt spray resistance. Cured ink impressions shall not deteriorate when exposed to a 20 percent salt spray solution at 33º to 37ºC for a period of not less than 48 hours.

3.9.4 Light fastness. Cured ink impressions shall not fade and shall remain legible when tested by a light fastness test. To determine conformance, one half of the surface of the test specimens, one glass, prepared in accordance with 3.9.4.1, and one copper, in accordance with 3.9.4.2, shall be covered to obscure light, and the remaining half shall be exposed for 24 hours to the radiation of a glass-enclosed, carbon-arc lamp in accordance with Method 5660 of FED-STD-191.

3.9.4.1 Glass specimen. Prepare test specimens by applying the ink to two 4 x 4-inch pieces of glass. Using a stencil brush conforming to CID A-A-2903, stencil the letters “ABCDE” approximately .750 inches high. Use sufficient ink to obtain neat, legible letters, but not enough to cause the ink to smear. Cure in a flat (horizontal) position (see 3.6 and 3.7).

3.9.4.2 Copper, aluminum, steel, and plastic specimens. Prepare test specimens by applying the ink to two 4 x 4-inch pieces of the following materials: sheet copper, aluminum, steel, and plastic. Using a stencil brush conforming to CID A-A-2903, stencil the letters “ABCDE” approx-imately .750 inches high. Use sufficient ink to obtain neat, legible letters, but not enough to cause the ink to smear.

3.9.5 Stability. Cured ink impressions shall not fade, chip, peel, or flow and shall remain legible when exposed to a temperature of 118º to ± 3ºC for a period of not less than 24 hours.

3.9.6 Fungus resistance. Cured ink impressions shall not support fungi growth when inspected. To determine conformance, cured ink impressions - one glass test specimen of 3.9.4.1 and one plastic test specimen of 3.9.4.2, shall be tested in accordance with MIL-STD-810, method 508.4.

3.9.7 Storage life. The ink compounds shall be free from grit, lumps, and skins and shall not gel, settle, harden, or otherwise deteriorate while stored for not less than one year in the original containers.

3.10 Workmanship. The materials shall be free of foreign particles and other contamination and shall be of uniform quality, condition, and appearance.

4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

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5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability (MA). The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The company producing the item must have been producing a product meeting the requirements of this CID for at least two years.

5.2.2 The company must have sold 1000 units meeting the requirements of this CID in the commercial marketplace over the past two years.

5.3 Inspection requirements. Bid samples may be required for CID items when necessary to ensure product quality.

5.3.1 Sampling for visual examination. Sampling for fill, color, workmanship, packaging, packing, and marking may be performed on one unit package.

5.3.2 Sampling for testing. Unless otherwise specified, the sample shall consist of one sample approximately equal in weight and volume to a unit package drawn from a single homogeneous mixture.

6. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

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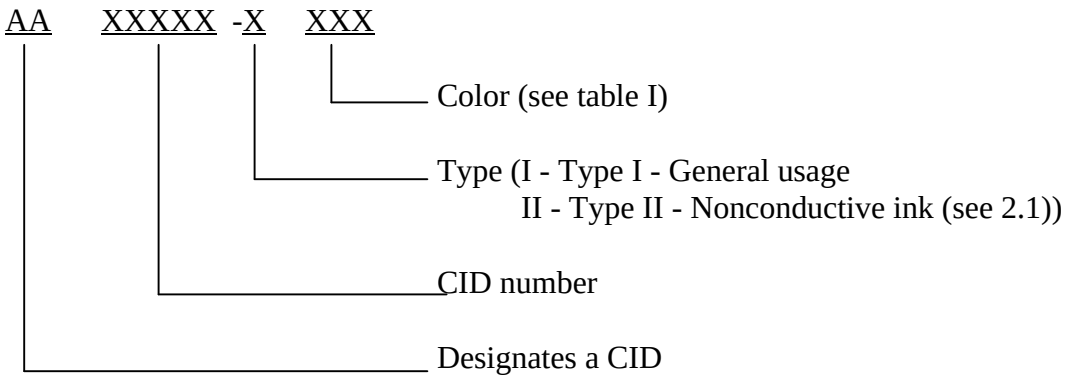
7. NOTES.

7.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for marking ink acquired to this CID is created as follows:



Example: AA56023-IIBLK (for Type II, Black)

TABLE I. Color identification.

Color	Dash identification character
Black	BLK
Red	RED
Orange	ORN
Yellow	YEL
Green	GRN
Blue	BLU
White	WHT

7.2 Handling precautions. Handling precautions shall be per manufacturer instruction.

7.3 Cross-reference data. Applications which reference MIL-I-43553, dated 12 January 1968, and which are in use prior to the approved date of Revision A, dated 25 November 1985, apply to Type I ink (general usage), which is not to be used for marking printed wiring boards.

7.4 Source of Documents.

7.4.1 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

7.4.2 Military Standards may be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.

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7.5 Product availability. This epoxy base marking ink is available commercially from Union Ink Company Inc, 453 Broad Avenue, Ridgefield, NJ 07657, and Haven Corporation, 6630 Quad Avenue,Baltimore, MD 21237.

7.6 Concluding material.

Custodian:
Army-MI
Navy - AS
Air Force - 99

Preparing Activity:
Army - MI

Project No. 7510-0412

Review Activities:
Army - GL,MD, AR, EA, AL, ME
Navy - OS
Air Force - 84

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